

Module3Lab3

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1 Module 3 Lab 3: Textbook Problem 2.2

1.0.1 Outliers in the Bridges Data

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Repo: <https://github.com/raulmckay-blip/cs82a-portfolio>

1.1 Overview

This lab looks through the `bridges` dataset for values that don't seem believable in its one numeric column, `length`. Unlike Lab 2 with the `USArrests` data, there's no clear rule here. A bridge length in feet doesn't have an obvious impossible range the way a percentage does.

```
[13]: import pandas as pd

df = pd.read_csv("bridges.csv")
print(df.shape)
df.head()
```

(108, 13)

```
[13]:
```

	Id	river	location	erected	purpose	length	lanes	clear-g	t-or-d	\
0	E1	M	3.0	1818	HIGHWAY	NaN	2.0	N	THROUGH	
1	E2	A	25.0	1819	HIGHWAY	1037.0	2.0	N	THROUGH	
2	E3	A	39.0	1829	AQUEDUCT	NaN	1.0	N	THROUGH	
3	E5	A	29.0	1837	HIGHWAY	1000.0	2.0	N	THROUGH	
4	E6	M	23.0	1838	HIGHWAY	NaN	2.0	N	THROUGH	

	material	span	rel-l	type
0	WOOD	SHORT	S	WOOD
1	WOOD	SHORT	S	WOOD
2	WOOD	NaN	S	WOOD
3	WOOD	SHORT	S	WOOD
4	WOOD	NaN	S	WOOD

1.2 Summary Statistics

```
[14]: df["length"].describe()
```

```
[14]: count      81.000000
      mean      1567.469136
      std       747.491523
      min       804.000000
      25%      1000.000000
      50%      1300.000000
      75%      2000.000000
      max      4558.000000
      Name: length, dtype: float64
```

1.3 My Assumption

Since I don't really know what counts as a normal bridge length in this dataset. It has highway, railroad, and aqueduct bridges built between 1818 and 1986. So I'm using a statistics based rule: a **length** more than **1.5 times the IQR above the 75th percentile** is unusual enough in this sample to flag for a second look.

```
[15]: q1, q3 = df["length"].quantile([0.25, 0.75])
      iqr = q3 - q1
      upper = q3 + 1.5 * iqr
      print(f"Flagging any length above {upper:.1f} ft")
      df[df["length"] > upper]
```

Flagging any length above 3500.0 ft

```
[15]:      Id river  location  erected  purpose  length  lanes  clear-g  t-or-d  \
31   E34      0      41.0    1888      RR   4558.0    2.0      G  THROUGH
44   E46      A      37.0    1897      RR   4000.0    2.0      G    DECK
104  E91      0      44.0    1975  HIGHWAY  3756.0    6.0      G  THROUGH

      material  span rel-l  type
31      STEEL  LONG      F  SIMPLE-T
44      STEEL  LONG      F  SIMPLE-T
104     STEEL  LONG      F    ARCH
```

1.4 Result

This rule flags **3 rows Id's**: E34, E46, and E91.

Before deciding what to do with them, I checked if anything else in the data backs up these values. The dataset has a separate column called **span**, with values **SHORT**, **MEDIUM**, or **LONG**. All three flagged bridges are also labeled **span = LONG**, and that label wasn't calculated from **length** at all, it's its own column. Two different columns agreeing is a good sign these are real long bridges and not typos.

```
[16]: df.loc[df["length"] > upper, ["Id", "river", "erected", "purpose", "length",  
    ↪ "span"]]
```

```
[16]:      Id river  erected  purpose  length  span  
31   E34     0    1888      RR  4558.0  LONG  
44   E46     A    1897      RR  4000.0  LONG  
104  E91     0    1975  HIGHWAY  3756.0  LONG
```

1.5 Deliverable Summary

My assumption: A `length` value more than 1.5 times the IQR above the 75th percentile, which comes out to above 3,500 ft here, is unusual enough in this sample to take a second look at.

Flagged rows: E34 (4,558 ft), E46 (4,000 ft), E91 (3,756 ft).

What I did: I kept all three values as they are. Each one is backed up by the dataset's own `span` column, which separately marks all three as `LONG`, a label that wasn't calculated from `length`. Since two different columns agree, I'm treating these as real bridges instead of mistakes. That's the same reasoning I used for South Carolina's assault rate in Lab 2. It's unusual but not impossible, and taking out real extreme values would hide real differences in the data instead of cleaning it up.

1.6 Note: Missing Values

Separate from everything above, `describe()` also shows that only 81 out of 108 rows actually have a `length` value. The other 27 are missing. That's not a bad or unbelievable value, it's just data that isn't there, so I'm not flagging or changing it. Still worth keeping in mind, since it means the length check above only covers about ***75 percent of the dataset***.